

DUNGEON KNIGHT

Game Design Document · v1.0 · Nurzhigit · Game Development Fundamentals, Narxoz University · Assignment 8-9

1 Title and pitch

Dungeon Knight: a knight holds a torch-lit dungeon room against endless slimes and bats, firing auto-aimed magic bolts until his three hearts run out.

2 Genre and platform

GENRE	PLATFORM	MODE
Arena shooter, roguelite-style	Android phones and tablets	Single player, landscape

Engine: Unity 6 (6000.0.23f1), built-in 2D renderer, uGUI, Input System. Inspired by Soul Knight. Session length is 2 to 5 minutes, sized for short mobile play.

3 2D or 3D

2D, top-down, three-quarter view. Sprites use 16 pixels per world unit and point filtering, so one floor tile equals one unit. Draw order follows each character's feet, so the knight walks in front of and behind pillars correctly. The camera fits the full 21 × 13 tile arena on any aspect ratio.

4 Core gameplay loop

Move to stay clear of enemies → Hold fire, bolts auto-aim → Kill: score + chance of a coin → Collect coins → Spawns speed up, bats join ↺

A run ends when the knight loses all three hearts. The game-over screen shows score and coins, with Try Again and Main Menu. After a hit the knight blinks and can't take damage for 1.2 seconds. Enemies spawn every 2.5 s at first, ramping down to every 0.9 s over 90 seconds, with at most 8 alive at once. They spawn only at points at least 4.5 units from the player and never inside walls.

5 Controls

Function	Touch (Android)	Keyboard (editor / PC)
Move, 8 directions	Virtual joystick, bottom-left (360 px, 520 px touch zone)	WASD or arrow keys
Shoot, hold to repeat	Fire button, bottom-right (260 px)	Space
Aim	Automatic: nearest enemy within 9 units, otherwise the facing direction	
Pause	Pause button, top-right (140 px)	Esc

The UI scales from a 1920 × 1080 reference matched on height. All touch targets are at least 120 px and all text at least 40 px. The HUD stays inside the screen's safe area on notched phones and keeps its layout on 4:3 tablets.

6 Content: levels, enemies, items

Type	Content in v1.0
Level	One hand-designed arena: stone walls with three torches on the top and bottom walls, pillars, crates and barrels as cover, six enemy spawn points.
Slime	3 HP, slow (1.7 u/s), 100 points, 45% coin drop. Available from the start.
Bat	2 HP, fast (3.0 u/s) with a wobbling flight path, 150 points, 30% coin drop. Joins after 20 seconds.
Magic bolt	1 damage, 11 u/s, fires every 0.28 s with slight spread, knocks enemies back, breaks on walls.
Coin	Pops out of a defeated enemy, flies to the player within 2.2 units, fades out after 25 seconds.

7 Art, sound and asset sources

Art: 16 × 16 pixel art in a dark purple-and-stone palette with warm torch light. Every sprite, UI frame and the bitmap-style font were drawn procedurally by Python scripts in the repository (`Tools/generate_art.py`, `Tools/generate_font.py`) for this project. Animation uses sprite flipbooks: idle, walk, enemy wiggle, spinning

coins, torch flames, hit sparks and death puffs. The game also has hit flash, screen shake, a vignette and flickering torch glow. No third-party or store assets are used.

Sound: none in v1.0. Planned: short chiptune effects for shoot, hit, coin and death, plus one looping dungeon track. Sources would be original jsfxr-style effects or CC0 packs with credit in the README.

8 MVP for the week-15 build

- done Main menu, dungeon HUD, pause menu, game over
- done Responsive layout: 16:9, notched phone, 4:3 tablet
- done Movement, auto-aim shooting, 2 enemy types, score, coins
- planned Sound effects and music, with a mute toggle
- planned Saved best score and total coins
- planned Third enemy type with a ranged attack
- planned Tested APK on a real Android device

9 Cut list

- Procedurally generated rooms and multiple floors
- Multiple weapons, inventory, shop
- Skill trees and character classes
- Multiplayer and online leaderboards
- Bosses with scripted attack patterns
- Ads, in-app purchases, accounts

10 Development plan, weeks 9-13

Week	Goal	Done when
9	**Sound.** Add an audio manager, effects for shoot / hit / coin / death / button, one music loop, a mute button in the pause menu.	Every action plays a sound, and mute persists between runs.
10	**Saving.** Store best score, total coins and the mute setting with PlayerPrefs. Show the best score on the main menu and game-over screen.	Values survive closing the app.
11	**Enemy AI.** Add a skeleton archer that keeps distance and shoots slow arrows. Make enemies steer around pillars instead of pushing into them.	Three enemy types, and no enemy stays stuck on cover.
12	**Effects and feel.** Damage numbers, coin pickup sound and burst, knight death animation, a short wave banner every 30 seconds.	Playtesters can tell when difficulty rises.
13	**Refinement.** Playtest with 3-5 classmates, balance spawn rates and HP, fix bugs, profile on a mid-range phone, build the release APK.	60 FPS on test device, known bugs fixed, APK ready for week 15.